



Kabadiwala Connect – Tech Stack

Frontend

Layer	Technology
UI Library	React 18
Language	TypeScript (strict mode, noUnusedLocals)
Build Tool	Vite
Styling	Tailwind CSS (custom design system: deep green / emerald / charcoal)

Routing	React Router DOM (role-guarded routes, SPA)
Charts	Recharts (earnings, analytics, waste trends)
Icons	Lucide React
Map	Custom SVG map visualization (Leaflet- swappable)
i18n	Custom translation architecture – English / हिन्दी / ગુજરાતી

Backend

Layer	Technology
Framework	Python FastAPI (40+ REST endpoints)
Validation	Pydantic v2
Server	Uvicorn (ASGI)
Uploads	python-multipart (type allowlist, 8 MB cap, safe filenames)
Auth	PBKDF2 hashing (120k iterations) + HMAC-signed session tokens + RBAC
Security headers	CSP, HSTS, X-Frame-

Options, Referrer-
Policy, Permissions-
Policy

Database

Layer	Technology
Cloud	Neon PostgreSQL (serverless, pooled connections)
Local demo	SQLite (zero-config, stdlib-only persistence)
Architecture	Dual-dialect — one codebase, auto-selected by

	DATABASE_URL, query translation layer
Schema	13 tables: users, collectors, recyclers, pickups, items, transactions, impact_records...

AI / Intelligence Layer (modular & replaceable)

Service	Approach
Waste Classifier	Demo inference engine — public API matches a real CV service

Value Estimator	Transparent reference-rate engine (₹/kg per material)
Collector Matcher	Weighted scoring: distance .40 · availability .25 · material .20 · rating .15
Route Optimizer	Nearest-neighbour heuristic with km/ time-saved output

Deployment

Layer	Technology

Hosting	Render — single service serves API + built SPA (same origin, no CORS)
Source control	GitHub (arbab-25/FALTU) with autodeploy on push
IaC	render.yaml Blueprint

Quality & Testing

Tool	Purpose
selftest.py	33-check API regression suite (runs on SQLite and Neon)

Bandit · pip-audit	Python security & CVE scanning
tsc · npm audit	TypeScript strictness & dependency CVEs
axe-core	WCAG 2.1 AA accessibility (0 violations across 10 pages)
pyflakes · vulture	Dead code detection

One-liner for judges: *"React 18 + TypeScript + Tailwind frontend, FastAPI + dual-dialect PostgreSQL/SQLite backend, four swappable AI services, deployed as a single Render service on Neon Postgres — fully offline-capable for the demo."*

Honest-label note (SIH requirement): the AI components are demo engines behind real-service interfaces, clearly labelled in the UI and README — not trained models.